

Project Executable Files

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete source code (all files and folders)	Yes — all code is in this folder (5. Project Development Phase)
2	README / Setup Guide with instructions to run the project	Yes — README.md in this folder
3	requirements.txt / dependency file	Yes — requirements.txt
4	Database schema / seed files (if applicable)	NA — CSV-based data, no database schema needed
5	Environment configuration file (.env.example or similar)	NA — no API keys or secrets required to run locally
6	Deployed application URL (if hosted)	No — runs locally; IBM Watson cloud deployment script (deploy_watson.py) is provided but not yet run live
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — web application
8	Test files and test results	Yes — see 6.Project Testing
9	Demo video or walkthrough (if required)	Yes — see 8.Project Demonstration

Step 2: File / Folder Structure

File / Folder	Purpose
app.py	Flask application — routes, feature builder, prediction logic
data/	Data generation, feature engineering, model training scripts, and the consolidated Jupyter notebook
models/	Saved model (best_model.pkl), scaler (scaler.pkl), feature schema (feature_columns.json), model info (model_info.json), and comparison charts
templates/	HTML pages: index.html (new application), result.html (decision), batch.html

	(screening), about.html (model report), base.html (shared layout)
static/	CSS design system (style.css), sample batch CSV (sample_batch.csv), and chart images
deploy_watson.py	IBM Watson Machine Learning deployment script (ready to run with IBM Cloud credentials)
requirements.txt	Python dependencies
README.md	Setup and run instructions, dataset swap guide, and cloud deployment notes

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not deployed publicly — runs on http://127.0.0.1:5000 locally
Login Credentials (Demo)	NA — no authentication in this build
Platform / Hosting	Local Flask development server (IBM watsonx.ai supported via deploy_watson.py, not yet run live)
Repository Link	https://github.com/siri-k08/Credit-Card-Approval-Prediction
Demo Video Link	https://drive.google.com/file/d/1OXnscJFOcufBZd1djGYXW252c_O4GDSE/view?usp=sharing

Step 4: Run Instructions

1. Unzip / clone the project and open the folder in VS Code or any editor.
2. Open a terminal in the project folder.
3. Run: `pip install -r requirements.txt`
4. Run: `python app.py`
5. Open `http://127.0.0.1:5000` in your browser.
6. Try New Application (single prediction), Batch Screening (upload static/sample_batch.csv), and Model Report.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Trained on synthetic data, not the real-world applicant dataset	Swap in the real Kaggle CSVs (same column names) and re-run the training pipeline — no code changes needed
2	IBM Watson Machine Learning deployment not yet live	Script is written and documented in <code>deploy_watson.py</code> ; requires the user's own IBM Cloud credentials to execute
3	No authentication / role-based access	Planned as future work; not required for the current single-team internal-tool use case